

Probability and Statistics

5

Unit Overview

In this unit, you will investigate relationships between two variables, and you will practice displaying, summarizing, and analyzing bivariate (two-variable) data. Using two numerical variables, you will investigate the *strength*, *form*, and *direction* of association between the two variables. Where appropriate, you will practice developing linear equations that model some of these relationships. For the case of two categorical variables, you will develop two-way tables that summarize the data in a way that allows for easy comparison between different categories. You will also develop graphical representations that can assist in the comparison of the data between different categories.

Key Terms

As you study this unit, add these and other terms to your math notebook. Include in your notes your prior knowledge of each word, as well as your experiences in using the word in different mathematical examples. If needed, ask for help in pronouncing new words and add information on pronunciation to your math notebook. It is important that you learn new terms and use them correctly in your class discussions and in your problem solutions.

Academic Vocabulary

- association
- deviate
- cluster

Math Terms

- association
- positive association
- negative association
- linear association
- non-linear association
- linear model
- bivariate data
- mean absolute deviation
- trend line
- two-way table
- categorical variables
- segmented bar graph
- row percentages

ESSENTIAL QUESTIONS



How does a scatter plot help you to investigate and interpret associations between two numerical variables?



How can the slope and y -intercept components of a linear model be interpreted in context when used to describe a linear association between two numerical variables?



How can a two-way table be used to assess an association between two categorical variables?

EMBEDDED ASSESSMENTS

These assessments, following activities 33 and 35, will give you an opportunity to demonstrate your understanding of bivariate association.

Embedded Assessment 1:

Scatter Plots, Associations, and Trends p. 465

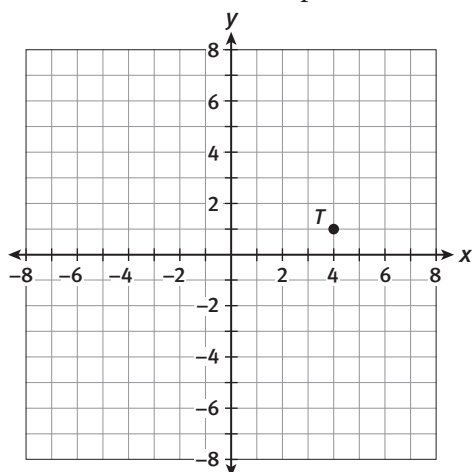
Embedded Assessment 2:

Median-Median Line and Two-Way Tables p. 485

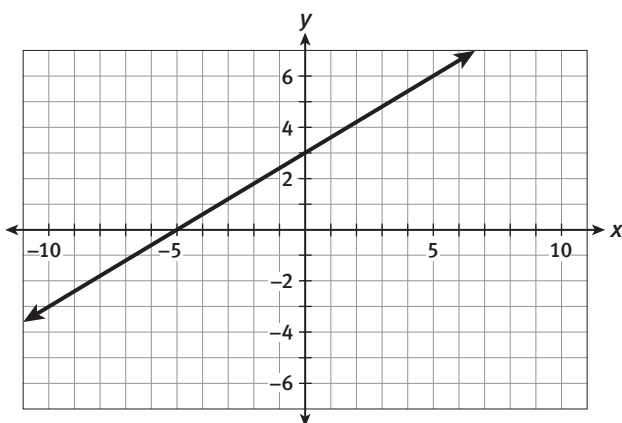
Getting Ready

Write your answers on notebook paper.
Show your work.

1. A line contains the points (2, 5) and (4, 6).
 - a. Where does it cross the x -axis?
 - b. Where does it cross the y -axis?
2. Use the graph below to
 - a. Plot and label the points $R(3, 5)$ and $S(6, 0)$.
 - b. Give the coordinates of point T .



3. Write the equation of the line graphed below.



4. Find the slope of the line containing these three points.

x	y
0	11
2	7.5
4	4

5. Write the equation of the line that contains the three points listed in Item 4. Write the equation in slope-intercept form.
6. State whether each of the following is an example of a numerical or categorical variable.
 - a. eye color
 - b. type of fruit
 - c. student height
 - d. textbook weight
 - e. favorite sport
7. Express the following fractions as percentages:
 - a. $\frac{15}{60}$
 - b. $\frac{42}{105}$
 - c. $\frac{146}{200}$
8. Convert the following two fractions to percentages (to the nearest percent) and determine which fraction is larger:

$$\frac{45}{92}, \frac{34}{65}$$

Analyzing Data

Cracker Snacker

Lesson 32-1 Scatter Plots

ACTIVITY 32

Learning Targets:

- Make a scatter plot.
- Recognize patterns in scatter plots.

SUGGESTED LEARNING STRATEGIES: Activate Prior Knowledge, Create Representations, Look for a Pattern, Think-Pair-Share

At a recent lunch, Mary was looking at the nutritional information printed on the packaging of some of the crackers she and her friends were eating. She noticed that some brands of crackers had much higher sodium than other brands, and that there was considerable variability in some of the other nutritional measurements (such as total fat). She also noticed that some of the brands that were high in one nutritional measurement were quite high or quite low in another. For example, one brand of crackers was quite high in both total carbohydrates and sodium, but it was fairly low in total fat. Mary wondered if some of the nutritional measurements for these crackers might be related.

Mary went to a large grocery that carries many brands of crackers and chose a random sample for which to record data. In a *random sample*, each member of the population (in this case, all the brands of crackers) has an equal chance of being chosen. Mary put all the different brands of crackers in a grocery cart and then closed her eyes and pulled nine packages from the cart. She recorded the following data for the nine brands of crackers, each based on a 30-gram serving.

Cracker (Serving = 30 g)	Total Fat (g)	Sodium (mg)	Fiber (g)	Total Carbohydrates (g)	Whole Wheat?
Fun Crisps	6	300	0	20	No
Fun Crisps— MultiGrain	7	250	1	18	No
Snax—Cheddar	5	350	1	20	No
Snax—Pretzel	3	450	1	24	No
Super Grain	3	250	4	23	Yes
Table Thin Crisps	3	300	1	22	No
Waves of Wheat	4	200	4	21	Yes
Zaps	8	250	0	19	No
Zaps—Low Salt	8	200	5	18	Yes

1. Describe any observations you have about the table.

My Notes

MATH TIP

Since it is often impossible to sample an entire population (think about the millions of people in a large city), a random sample can be used to estimate the traits of the entire population.

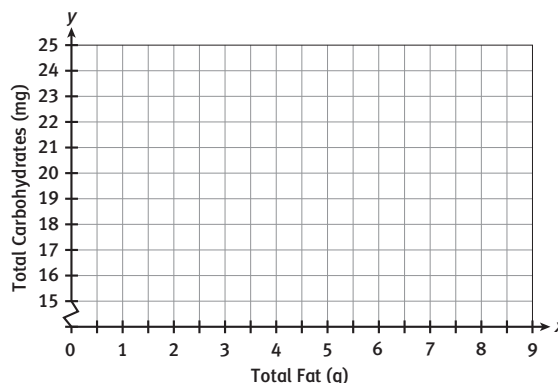
My Notes

ACADEMIC VOCABULARY

Association is the degree to which two variables are related, generally described in terms of the association's direction, form, and strength.

2. Examine the **association** between two numerical variables by creating scatter plots.

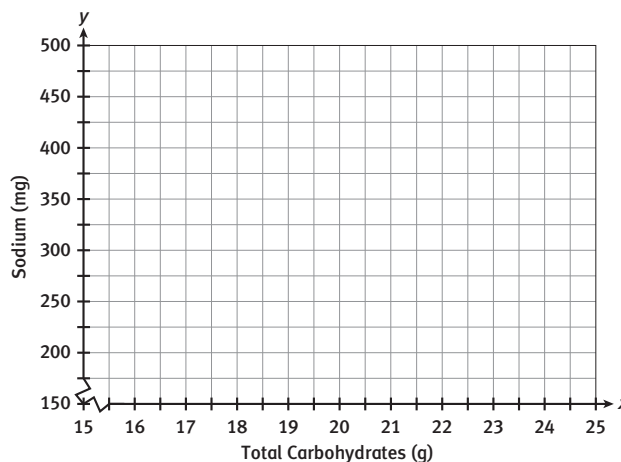
- a. **Model with mathematics.** Make a scatter plot for the nine brands of crackers, recording total fat on the horizontal axis and total carbohydrates on the vertical axis. Place a circle around any point on the graph that represents a whole-wheat brand of cracker.



- b. Describe any patterns you observe from the scatter plot.

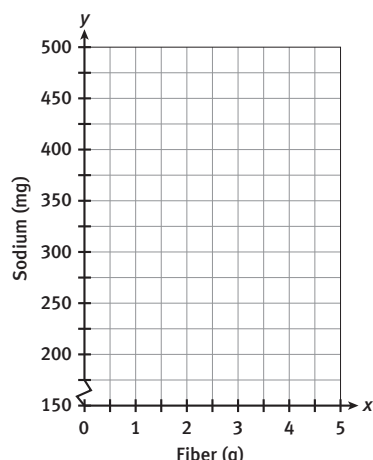
3. Look at the association between total carbohydrates and sodium.

- a. Construct a scatter plot for the nine brands of crackers, recording total carbohydrates on the horizontal axis and sodium on the vertical axis. As before, place a circle around any point on the graph that represents a whole-wheat brand of cracker.



- b. Do the whole-wheat crackers generally have higher or lower sodium than the other crackers?

4. Make a scatter plot to compare fiber and sodium for the nine brands of crackers.
 - a. Record fiber on the horizontal axis and sodium on the vertical axis. Again, place a circle around any point on the graph that represents a whole-wheat brand of cracker.



- b. Are the fiber amounts in the whole-wheat crackers very different from the fiber amounts in other crackers? How can you tell from the graph?

Check Your Understanding

5. When collecting statistical data, what is the value of using a random sample versus a sample of an entire population?
6. **Critique the reasoning of others.** Compare your three scatter plots with a classmate's, and if you want to make any changes or improvements to your graphs, do so now. Keep in mind that relationships and patterns in scatter plots need not always be lines.
7. Consider the scatter plot you constructed in Item 3. Which of the following statements (A or B) describes the pattern in this scatter plot?

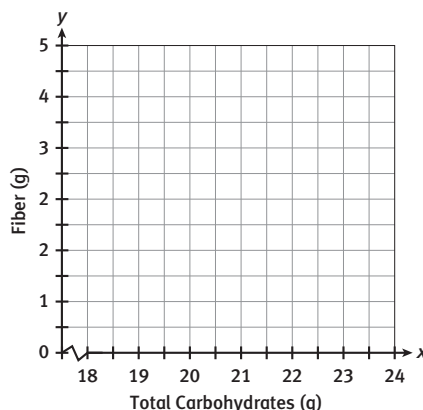
Statement A: Crackers that have more grams of total carbohydrates tend to have less sodium than crackers with less total carbohydrates.

Statement B: Crackers that have more grams of total carbohydrates tend to have more sodium than crackers with less total carbohydrates.

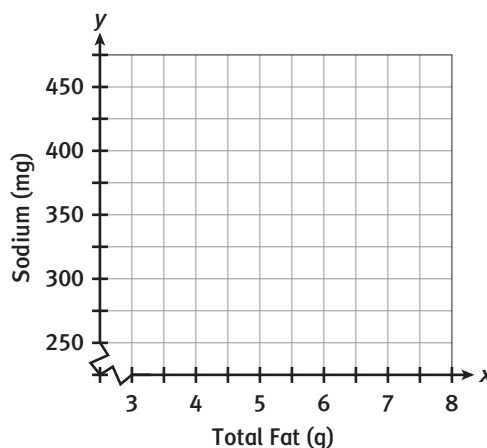
My Notes

LESSON 32-1 PRACTICE

8. Explain how you would generate a random sample, and give an example.
9. Use the cracker data to construct a scatter plot with total carbohydrates on the horizontal axis and fiber on the vertical axis.



10. Notice that in the scatter plot from Item 7, three points stand out as different from the others. What do the crackers that correspond to these three points have in common?
11. Use the data for the six crackers that are not whole-wheat to construct a scatter plot that uses sodium on the vertical axis and total fat on the horizontal axis.



12. **Construct viable arguments.** Based on the scatter plot in Item 9, describe the relationship between total fat and sodium content for these crackers.

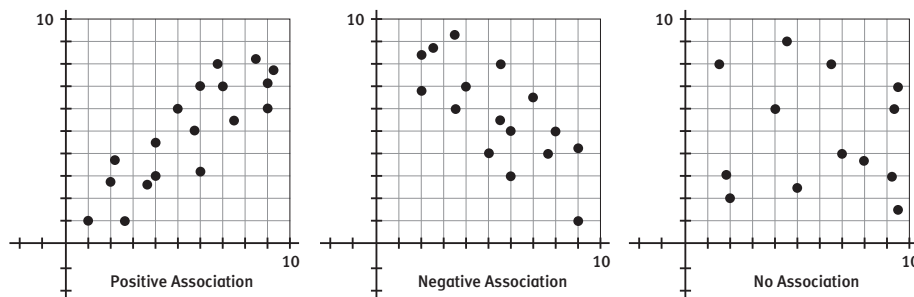
Learning Targets:

- Recognize patterns in scatter plots.
- Describe association between two numerical variables in terms of direction, form and strength.

SUGGESTED LEARNING STRATEGIES: Marking the Text, Close Reading, Look for a Pattern, Visualization, Summarizing, Paraphrasing

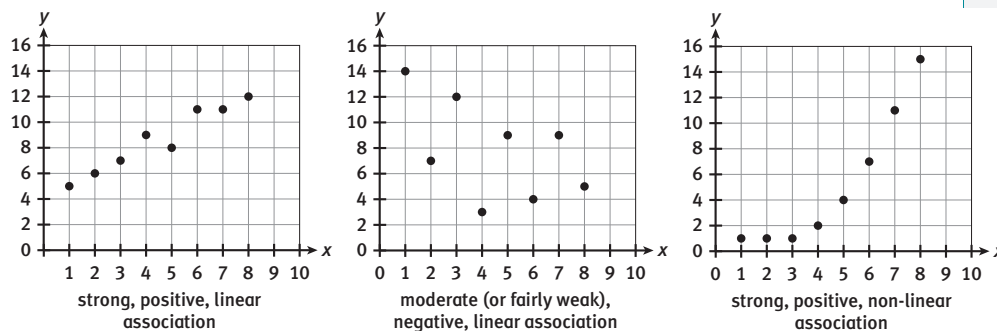
Direction, form, strength, and any other special characteristics are considered when describing the association of data in a scatter plot. As you continue reading, be sure to mark or highlight the text to identify words and phrases that help you understand the context and key terms.

Direction: Two variables have a **positive association** if y tends to increase as x increases. They have a **negative association** if y tends to decrease as x increases. If the data have no clear relationship, they have **no association**.



Form: If the data points in a scatter plot generally follow a linear pattern, the relationship is described as a **linear association**. If the collection of data points follows any type of noticeably curved pattern, the relationship is described as a **non-linear association**. It is also possible that there is no clear relationship at all.

Strength: When points fall close to an imaginary line or a curve, the association is considered to be **strong**. The more closely the points tend to follow the path of a line or curve, the stronger the association is. On the other hand, if the points are somewhat widely scattered around an imaginary line or curve, the association is considered to be **weak**. A **moderate** association is one that is neither strong nor weak, but “in between.”



My Notes

MATH TERMS

Two variables are related in a **positive association** when values for one variable tend to increase as values for the other variable also increase; an increase in x corresponds with an increase in y .

Two variables are related in a **negative association** when values for one variable tend to decrease as values for the other variable increase; an increase in x corresponds with a decrease in y .

When a linear pattern (such as one of the form $y = mx + b$) describes the essential nature of the relationship between two variables, they have a **linear association**.

When a non-linear pattern (such as a curve) describes the essential nature of the relationship between two variables, they have a **non-linear association**.

CONNECT TO AP

In AP Statistics, you will use a statistic called the **correlation coefficient** to describe the strength and direction of a linear association.

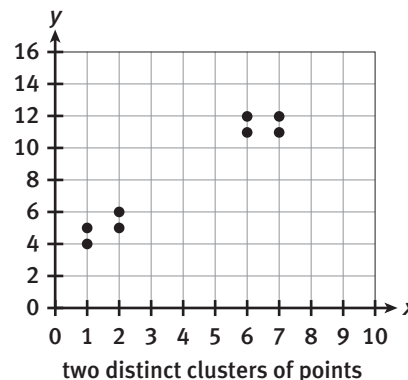
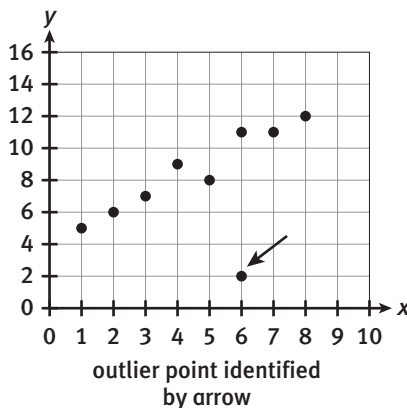
My Notes

ACADEMIC VOCABULARY

To **deviate** means “to stray or depart from an established course” or “to digress from a line of thought or reasoning.”

A **cluster** is a small group or bunch of something, like points in this case.

Other Special Characteristics: Any points that substantially **deviate** from the overall pattern, called **outliers**, are worth noting. Also, anytime a noticeable concentration of points is present, that **cluster** should be noted.



If more data were added to the second graph above for values of x from 3 to 5, they might reveal a strong linear relationship, but they might also reveal some other kind of relationship. Always be careful about inferring too much about data that are not present in your graph!

1. Complete the table below for each of the cracker data scatter plots you constructed in Items 2–4 in Lesson 32-1. State *none* for any table entry that is not applicable.

Graph	Direction	Form	Strength	Special/Other
Total Carbohydrates vs. Total Fat				
Sodium vs. Total Carbohydrates				
Sodium vs. Fiber				

2. **Reason abstractly.** Which of the three relationships above would be *best* suited for creating a **linear model** to describe the relationship? Explain your reasoning.
3. Which of the following linear models appears reasonable for describing the relationship between Total Carbohydrates (y) and Total Fat (x) based on your scatter plot?
- $y = 25 - x$
 - $y = 25 + x$
 - $y = 50 - 2x$

Check Your Understanding

4. Compare your descriptions from your table in Item 1 with a classmate. Describe any changes you would make.
5. **Reason quantitatively.** How did you choose your answer in Item 3? How did you determine which y -intercept and slope values would be the best choices?

LESSON 32-2 PRACTICE

6. Below is a data table showing the weights of seven newborn horses and the weights of the mother horses. Use the data to construct a scatter plot with mother's weight on the horizontal axis and newborn's weight on the vertical axis.

Newborn Weight (kg)	Mother Weight (kg)
130	550
120	650
130	600
120	550
125	625
125	650
115	575
120	600

My Notes

MATH TERMS

A **linear model** of the form $y = mx + b$ describes the essential nature of the relationship between y and x .

My Notes

7. Based on the scatter plot you constructed in Item 6, is there a relationship between newborn weight and mother weight? Justify your response.
8. To make chips, tortillas are fried in oil. Below is a data table on frying time (in seconds) and moisture content (percent) for eight chips. Construct a scatter plot with frying time on the horizontal axis and moisture content on the vertical axis.

Frying Time (seconds)	Moisture Content (%)
5	14
10	10
15	8
20	9
25	3
30	6
45	2
60	1

9. Does the relationship between frying time and moisture content look linear or non-linear?
10. **Construct viable arguments.** If moisture content is high, the chip can be soggy. Based on what you see in the scatter plot you constructed in Item 8, do you think it would be better to fry chips for 10 seconds or for 50 seconds? Justify your response.
11. Would you describe the relationship between frying time and moisture content as positive or negative?
12. Would you describe the relationship between frying time and moisture content as strong or weak?
13. Which of the following two lines would be better for describing the relationship between frying time and moisture content: $y = 8 + 0.5x$ or $y = 12 - 0.2x$? Justify your choice.

ACTIVITY 32 PRACTICE

Lesson 32-1

The following data were collected for a group of students.

Hours of TV per Week	Percent of Homework Completed	Test Score
32	58	66
13	82	85
28	65	75
19	87	85
11	98	100
21	78	88
15	75	85
11	92	90
15	75	90
12	91	95
17	90	85
20	81	85

- Construct a scatter plot that uses Hours of TV per Week on the horizontal axis and Percent of Homework Completed on the vertical axis.
- Construct a scatter plot that uses Percent of Homework Completed on the horizontal axis and Test Score on the vertical axis.
- Construct a scatter plot that uses Hours of TV per Week on the horizontal axis and Test Score on the vertical axis.
- A consumer group rated the quality of nine brands of bicycle helmets. The quality ratings and the prices for these nine brands are shown below. Construct a scatter plot that uses Price on the horizontal axis and Quality Rating on the vertical axis.

Quality Rating	Price
65	35
60	30
55	40
47	23
47	30
43	18
41	28
40	20
32	25

Lesson 32-2

- Which of the two scatter plots (from Items 1 and 2) has a positive association? How would you describe the strength and form of this association?
- In the scatter plot of Test Score vs. Percent of Homework Completed, are there any interesting or unusual characteristics of the scatter plot or any unusual points?
- In the scatter plot you constructed in Item 3, would you describe the relationship between Test Score and number of Hours of TV per Week as positive or negative?
- Are there any interesting or unusual points or characteristics in the scatter plot you constructed in Item 3?

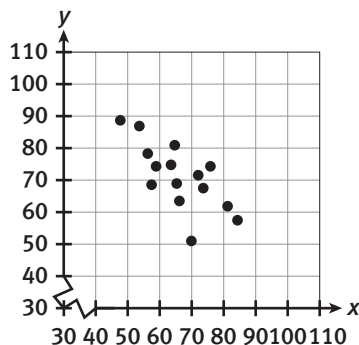
Use the scatter plot you constructed in Item 4 to answer Items 9–13.

- Does the relationship between Price and Quality Rating look linear or non-linear?
- Do bike helmets that cost more tend to have a higher quality rating?
- Would you describe the relationship between Price and Quality Rating as positive or negative?
- Would you describe the relationship between Price and Quality Rating strong, moderate or weak?
- Which of the following two lines would be better for describing the relationship between Price and Quality Rating: $y = 7 + 1x$ or $y = 5 + \frac{1}{2}x$? Justify your choice.

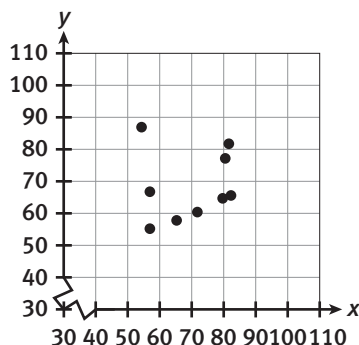
14. For each of the scatter plots shown, answer the following questions.

Does there appear to be a relationship between x and y ? If so, does the relationship appear to be linear? If so, would you describe the linear relationship as positive or negative?

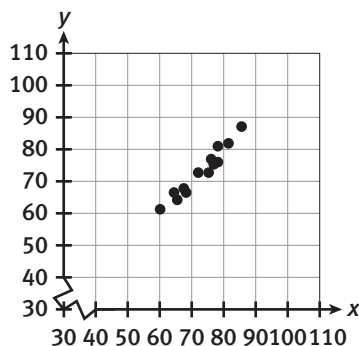
Scatter Plot 1



Scatter Plot 2

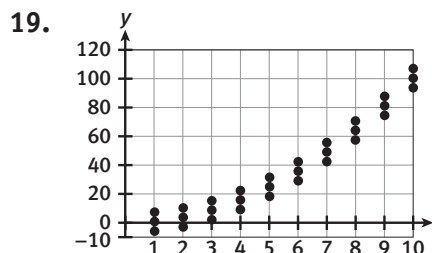
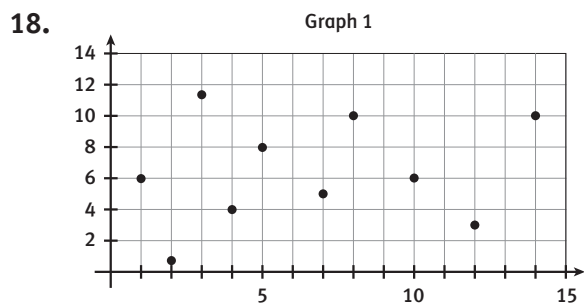
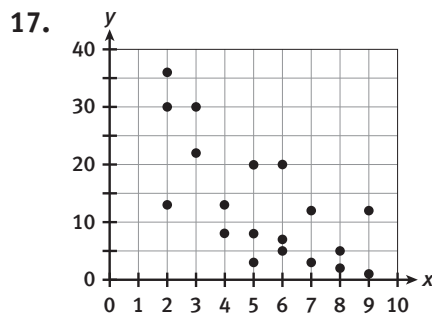
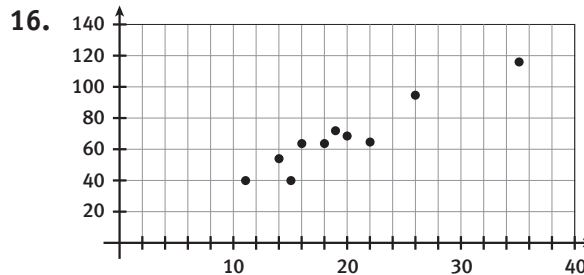


Scatter Plot 3



15. For each of the following pairs of variables, indicate whether you think there would be a relationship between these variables, and if so, whether it would be positive or negative.
- time spent studying for a test and score on the test
 - a person's weight and the time it takes the person to run one mile

For each of the scatter plots in Items 16–19 below, indicate whether the graph displays a strong positive, moderate positive, strong negative, or moderate negative association, or no association at all.



20. Which of the graphs in Items 16–19 demonstrate a linear association?
21. Describe the association between sodium and fiber in the crackers in Lesson 32-1, Item 4.

MATHEMATICAL PRACTICES

Use Appropriate Tools Strategically

22. Explain how a scatter plot can be used to describe the relationship between two variables.

Bivariate Data

Sue Swandive

Lesson 33-1 Collecting Data

ACTIVITY 33

Learning Targets:

- Collect bivariate data from an experiment.
- Summarize bivariate data in a scatter plot.

SUGGESTED LEARNING STRATEGIES: Activating Prior Knowledge, Think-Pair-Share, Self Revision/Peer Revision, Visualization

The famous bungee jumper, Sue Swandive, is promoting her new doll line. There will be a bungee competition with the new doll. The winner gets a special prize.

The competition rules are as follows:

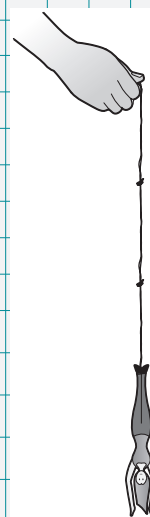
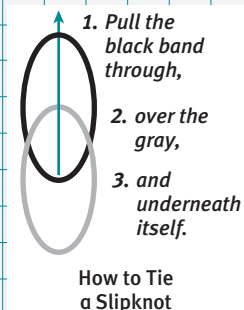
- Attach a rock (or other weighted item) using tape to the back of a Sue Swandive doll.
- Make a bungee cord by connecting rubber bands and attach it to the doll. (See diagrams at right. Secure the doll's feet together using one rubber band around the ankles. This will be used to attach the first rubber band with a slipknot.)
- Prepare to drop the doll from an appropriate height with the bungee cord attached. Before dropping the doll, predict how far the doll will drop.
- The winner(s) will be determined according to whose *prediction* for how far the Sue doll will drop is closest to the *actual* distance the Sue doll drops.

To help you make a prediction for the competition, you will collect data in your classroom first.

Begin the classroom part of your experiment as follows:

- Attach the bungee to the Sue doll. Hold the end of the bungee cord *and* the Sue doll's feet at the 0 position on the tape measure with the doll's head pointing straight down.
- Let go of the doll's feet *but* hold onto the bungee cord.
- Record the height of the doll's head at its lowest position. (It may be helpful to tie the doll's hair back.)
- Be prepared to repeat each jump a few times to get a representative measurement. (You might want to try each jump at least three times and record the *median* measurement from your multiple attempts using a bungee cord that has a given number of rubber bands.)
- Add rubber bands to extend the length of the bungee cord and continue to take readings. Use slipknots to attach the rubber bands.

My Notes



My Notes

- Record the data for your class in the table below. Measure the Length of Bungee Jump from the 0 point of the tape measure.

Number of Rubber Bands Attached to the Sue Doll	Length of Bungee Jump
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

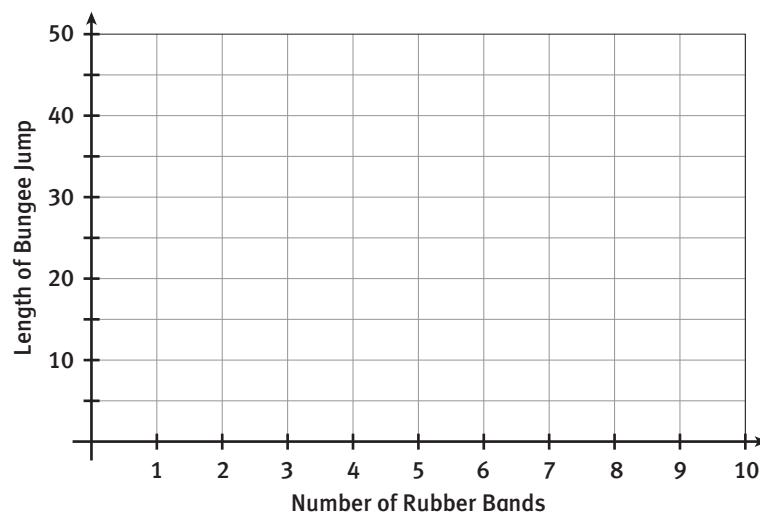
MATH TERMS

Bivariate data contains two variables for each individual, subject, observation, trial, etc.

The data you have recorded is an example of **bivariate data**.

Bivariate data are data that list two variables for each subject or observation (for example, number of rubber bands *and* length of jump, a person's height *and* weight, someone's favorite food *and* state of residence, and so on.)

- Attend to precision.** Create a scatter plot of the data on the grid below. Adjust the scale of the y -axis according to your unit of measurement and the number of your rubber bands.



Collecting Data

ACTIVITY 33

continued

3. Do the data indicate a linear relationship between the number of rubber bands and the length of bungee jump? Explain your answers using both the scatter plot and the table.
4. Describe how the length of the bungee jump changes as the number of rubber bands increases. What are the direction and strength of this association?

A scatterplot displays data so that you can visually see the distribution of the data. In earlier grades, you have calculated the mean, median, and mode of data as measures of variability in the data. Another measure of the spread of a distribution is the **mean absolute deviation** (MAD). The MAD is the average distance that the observations are from the mean of the distribution. For example, if the mean of the bungee-jumping results for your class is 18 inches and your result was 22 inches, then your result is a distance of 4 from the mean—the distance between 18 and 22.

Assume that 10 of your classmates had the following distribution of results for the bungee jumping contest (in inches).

12 18 22 14 17 18 26 29 23 21

5. What is the mean for this distribution?
6. Create a table showing the results above. List the results in one column, and then calculate the distance from the mean for each data point in the second column. Add the distances from the mean to find a total.
7. Now find the mean absolute deviation (MAD) by finding the average, or mean, of the distances in the table.
8. Look back to Item 2 and the data you plotted on a scatter plot for the length of each bungee jump. Notice how the lengths are spread and relate this to the MAD.

My Notes

My Notes

Check Your Understanding

9. Compare your scatter plot and table with the scatter plots and tables of other classmates. How are the scatter plots and tables similar? How are they different? Explain your reasoning.
10. Describe the mean absolute deviation. How does it relate to the mean of a sample of data?

LESSON 33-1 PRACTICE

11. Four data sets are described below. Which of these data sets are bivariate data?

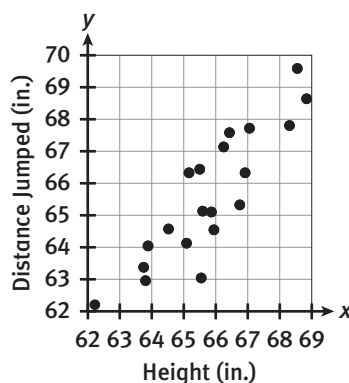
Data Set 1: Heights of 20 third graders.

Data Set 2: Prices and sizes of 40 houses.

Data Set 3: Amounts of fat and numbers of calories for 10 brands of hot dogs.

Data Set 4: Times to run 1 mile for 12 runners.

12. For each data set in Item 6 that is bivariate data, would you expect the association between the two variables to be positive or negative? Explain your choices.
13. **Make use of structure.** The scatter plot below displays data on height (in inches) and distance jumped (in inches) for twenty high school girls. How would you describe the association between height and distance jumped?



14. **Reason abstractly.** Imagine that you collected data for a new bungee jump experiment and that you did not add a weight to the doll in this experiment. If you were to make a scatter plot for the data from this new experiment, how do you think it would be different from the scatter plot you made in Item 2?
15. Calculate the MAD for the following data points: 22, 34, 21, 12, 40, 37, 27, 19, 23, 25.

Learning Targets:

- Informally fit a line to bivariate data.
- Use a trend line to make a prediction.

SUGGESTED LEARNING STRATEGIES: Activating Prior Knowledge, Think-Pair-Share, Self Revision/Peer Revision, Visualization.

A **trend line** is a line that indicates the general pattern in linearly associated data.

- 1. Use appropriate tools strategically.** Place a straightedge like a piece of cardboard or a ruler on the scatter plot in Item 2 in Lesson 33-1 in a position that would seem to indicate the general trend of the data and serve as a good, representative linear model. Trace that linear model on the coordinate grid. The line *does not* have to go through any of the data points you collected. It just needs to represent the general trend of the data.
- 2.** Write an equation in slope-intercept form for the trend line you just created.
- 3.** Explain what the variables in the equation of your trend line represent.
- 4. Make sense of problems.** How would you interpret the slope of your trend line in relation to the variable names you used above?
- 5.** In preparation for the competition, make a prediction for the distance your doll will drop based on the number of rubber bands your teacher specifies.

Prediction:

My Notes

CONNECT TO AP

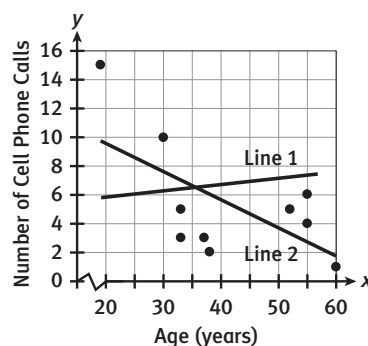
In AP Statistics, you will find trend lines for bivariate numerical data using a method called **Least Squares Regression**.

Check Your Understanding

6. Compare your trend line equation with the equations of other classmates. Why would the slopes be similar? Why would they be different?
7. Briefly explain how you chose your trend line model. What characteristics of the data were important to you in determining your equation?

LESSON 33-2 PRACTICE

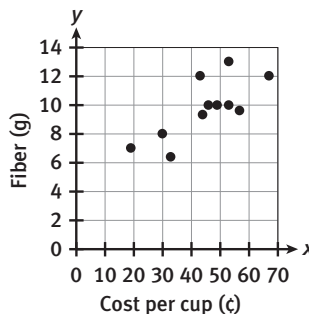
8. The scatter plot below was constructed using data on age and number of cell phone calls made in a typical day for 10 people. Two trend lines have been added to the scatter plot. Which line does a better job of describing the relationship between age and number of cell phone calls made?



9. **Make sense of problems.** The trend line $y = 492 + 15x$ was determined using data on age, x , and distance walked (in meters), y , for six minutes for boys between the ages of 3 and 18.

- a. Interpret the slope of the trend line.
- b. Based on this trend line, how many meters would you predict a 12-year-old boy would walk in six minutes?

10. The scatter plot below was made using data on x = cost per cup (in cents) and y = amount of fiber per cup (in grams) for 11 breakfast cereals. Graph a trend line on the scatter plot that you think does a good job of describing the relationship between fiber content and cost.



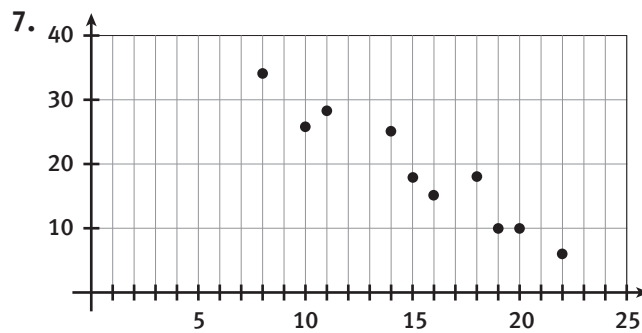
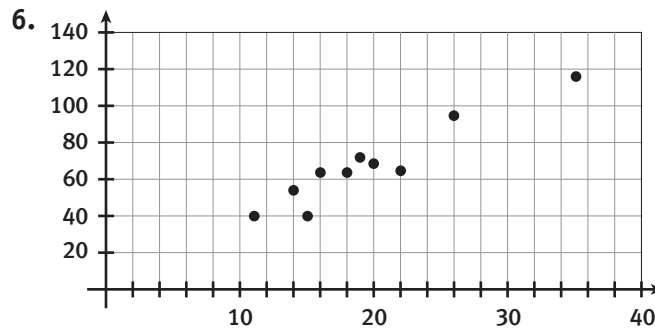
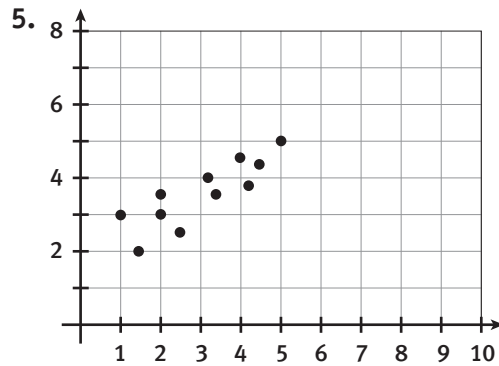
11. Write an equation for the trend line you drew in Item 10.
12. Based on your trend line model from Item 11, how many grams of fiber would you predict for a cereal that costs 40 cents per cup?

My Notes

Check Your Understanding

Write your answers on notebook paper. Show your work.

Each of the scatter plots in Items 5, 6, and 7 displays data that show an approximate linear trend. Write an equation for each trend line.



Lesson 33-3

The Competition!

ACTIVITY 33

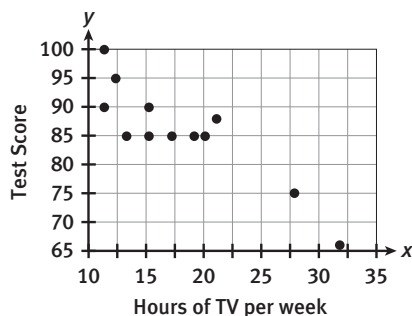
continued

For each trend line equation in Items 8 and 9, briefly interpret the slope in relation to the numerical variables.

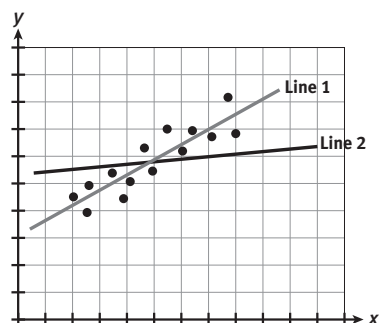
8. $y = 75 + .19x$, where x = number of miles traveled and y = the cost of an airline flight in dollars.
9. $y = 100 - 1.2x$, where x = number of hours of TV watched per week and y = test score on last week's test.
10. **Make sense of problems.** What does the direction of the association for a set of bivariate numerical data indicate about the slope of the corresponding trend line?

LESSON 33-3 PRACTICE

11. The scatter plot below shows the data for hours of TV per week and test scores.



- a. Graph a trend line on the scatter plot.
- b. Write an equation of the trend line you drew.
12. Do you think every student in your class will have the same equation or trend line in Item 11? Why or why?
13. **Critique the reasoning of others.** In the scatter plot shown below, explain why Jacinda's Line 1 does a better job of describing the trend than Lucas' Line 2.



ACTIVITY 33 PRACTICE

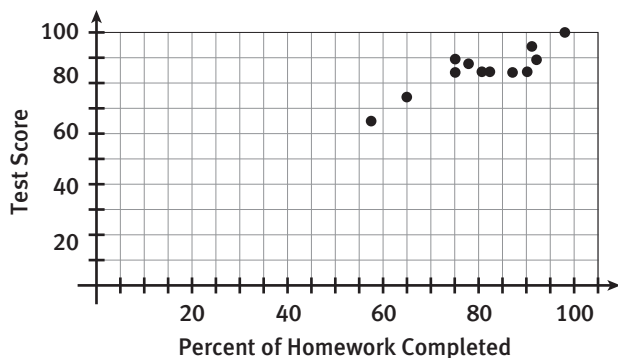
- Three data sets are described below. Which of these data sets are bivariate?

Data Set 1: Heights and weights of 17 bears.

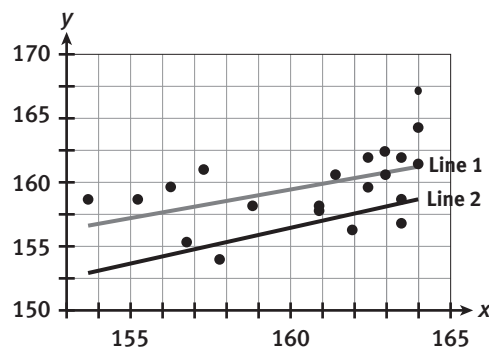
Data Set 2: Test scores for 25 students.

Data Set 3: Scores on a reading test and ages for 50 elementary school children.

- For each data set in Item 1 that is bivariate data, would you expect the association between the two variables to be positive or negative? Justify your response.
- The trend line $y = 12 - 0.2x$ was determined using data on x = frying time (in seconds) and y = moisture content (in %) for tortilla chips.
 - Interpret the slope of the trend line.
 - Based on this trend line, what moisture content would you predict for a frying time of 40 seconds?
- The scatter plot below was made using the data from Lesson 32-1 on x = percent homework completed and y = test score for 12 students. Graph a trend line on the scatter plot that you think does a good job of describing the relationship between test score and percent of homework completed.



- Write an equation of the trend line you graphed in Item 4.
- Based on your trend line equation from Item 5, what test score would you predict for a student who completed 80% of the homework?
- Two trend lines are shown on the scatter plot below. Which line does a better job of describing the relationship between x and y ?



- Explain why the line you chose in Item 7 is better than the other line.

MATHEMATICAL PRACTICES

Model with Mathematics

- Frances has planted a new African violet and, as time passes, new leaves are appearing. She has recorded this information in the table below. Create a linear equation that best models Frances's data and can be used to predict the number of leaves Frances should expect after 60 weeks. Let t represent the time (in weeks) and n represent the number of leaves on the plant.

Time in weeks	1	3	6	9	12	15	18	21
Number of leaves	1	2	3	3	4	6	7	9

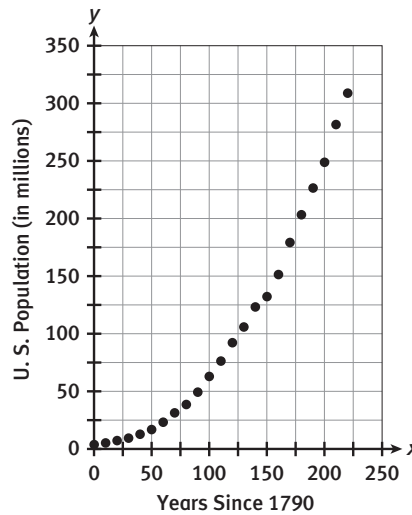
Scatter Plots, Associations, and Trends

U.S. CENSUS

Embedded Assessment 1

Use after Activity 33

1. The scatter plot below shows the U.S. population (in millions) from 1790 to 2010 according to data from the U.S. Census Bureau.



MATH TIP

A **census** occurs when the information of interest is collected on all members of a population. Collecting the survey information from all students is a census.

- a. What is the *direction* of the association between U.S. Population and Years Since 1790? Briefly state a few characteristics of the data or the graph that support your description.
- b. What is the *form* of the association between U.S. Population and Years Since 1790? Is it linear or non-linear?
- c. What is the *strength* of the association between U.S. Population and Years Since 1790? State a characteristic of the graph that supports your description.
- d. Do you think a linear model would be a good model to use to explain the association between U.S. Population and Census Year? Why or why not?

2. Mr. Mokher and his college students went to the local elementary school to collect data. The names of all the students were written on slips of paper and placed into a large container, which was then shaken vigorously. Mr. Mokher and his college students drew 12 slips of paper, one at a time, from the container. After taking a random sample of 12 children from Grade 0 (Kindergarten) through Grade 6, they collected data on each child's grade level and height in inches.

Student	Grade	Height (in.)
1	0	36
2	0	38
3	0	40
4	1	39
5	2	40
6	2	42
7	3	45
8	4	47
9	5	47
10	5	50
11	5	60
12	6	53

- a. Construct a scatter plot of height (y -axis) versus grade (x -axis).
- b. Describe the overall association between the grade level and the student's height in inches in terms of direction, form and strength.
- c. Graph a trend line on your scatter plot.
- d. Write an equation for the trend line.
- e. Interpret the slope of your trend line in the context of this analysis.
- f. Identify any student who substantially deviates from the overall pattern. Describe in context how the student does not fit the overall pattern.
- g. Given the context of the data set and the trend line model you have created, do you think your model will be applicable for later grades (e.g., middle school, high school, college)? Justify your response.

Scoring Guide	Exemplary	Proficient	Emerging	Incomplete
	The solution demonstrates these characteristics:			
Mathematics Knowledge and Thinking (Items 1a-d, 2a-g)	- Clear and accurate understanding of associations in bivariate data. - Clear and accurate understanding of scatter plots and trend lines.	- Recognition of associations in bivariate data. - A functional understanding of scatter plots, and trend lines.	- Partial recognition of associations in bivariate data. - Partial understanding of scatter plots, and trend lines.	- Little or no understanding of associations in bivariate data. - Little or no understanding of scatter plots, and trend lines.
Problem Solving (Item 2g)	Clear and accurate interpretation of data displays to make a prediction.	Interpreting data displays to make a prediction.	Difficulty making an accurate prediction from data displays.	Inaccurate interpretation of data displays.
Mathematical Modeling / Representations (Items 1d, 2a, 2c-d)	- Clear and accurate understanding of linear and non-linear models. - Clear and accurate understanding of plotting data, drawing a trend line, and writing an equation from a trend line.	- Understanding the difference between linear and non-linear models. - Mostly correct plotting of data, drawing a trend line, and writing an equation from a trend line.	- Partial understanding of linear and non-linear models. - Partial understanding of plotting data, drawing a trend line, and writing an equation from a trend line.	- Little or no understanding of linear and non-linear models. - Inaccurate or incomplete understanding of plotting data, drawing a trend line, or writing an equation.
Reasoning and Communication (Items 1a, 1c, 2b, 2e-g)	- Precise use of appropriate math terms and language to characterize associations in bivariate data using a scatter plot or trend line. - Making clear and accurate predictions from a graph.	- Correct characterization of associations in bivariate data using a scatter plot or trend line. - Making reasonable predictions from a graph.	- Misleading or confusing characterization of associations in bivariate data using a scatter plot or trend line. - Making partially correct predictions from a graph.	- Incomplete or inaccurate characterization of associations in bivariate data using a scatter plot or trend line. - Making incomplete or inaccurate predictions from a graph.

Median-Median Line

ACTIVITY 34

Homework Help Line

Lesson 34-1 Finding the Median-Median Line

Learning Targets:

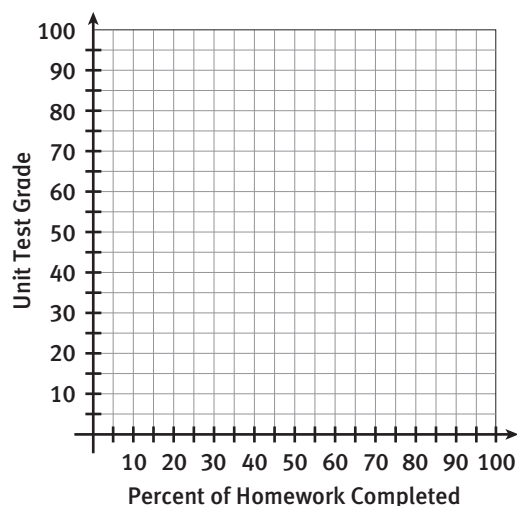
- Determine if a linear model is a good fit for a scatter plot.
- Find the median-median line for bivariate numerical data.

SUGGESTED LEARNING STRATEGIES: Activate Prior Knowledge, Think-Pair-Share, Quickwrite

Ms. Windle wants to examine the relationship between a student's percent of homework completed and the grade earned on the unit test. Below is a table showing the percent of homework completed and the earned test grade for each student in her class.

Percent of Homework Completed	Unit Test Grade
26	39
29	54
42	61
46	50
52	53
56	62
63	76
70	77
71	64
74	83
82	66
87	70
92	87
95	78
100	91

1. Let x represent the percent of homework completed and y represent the unit test grade. Construct a scatter plot of the data.



My Notes

My Notes

2. Make use of structure.

a. Describe the overall association between the two variables in terms of direction, form, and strength.

b. Why might Ms. Windle want to display the scatter plot to her students?

3. Would a linear model such as a trend line be appropriate here? If so, how well do you think the data would fit such a model? Explain.

4. If each student in your class were to fit a trend line by drawing a line on the scatter plot, would everyone get the same line?

Check Your Understanding

Imagine a scatter plot where you decide that a linear model will **not** fit the data well. Consider the characteristics of such a scatter plot.

5. What kind of form in a scatter plot would lead you to say that a linear model will *not* fit the data well?

6. Even with a linear form of association, what kind of strength would lead you to say that a linear model will *not* fit the data well?

When there is a strong linear association visible in a scatter plot, a trend line developed by visual examination tends to be a very good model. However, it would be better if we had an effective, specific procedure for developing such a model rather than just relying on visual judgment.

A **median-median line** can be used to describe and analyze bivariate data. Ms. Windle likes to use this modeling method because it is convenient to use with smaller data sets. Also, compared to other linear modeling methods, calculating the line's slope and y -intercept is typically not as influenced by outliers.

Lesson 34-1

Finding the Median-Median Line

ACTIVITY 34

continued

7. Keeping the x -coordinates of the points in increasing order, use the x -coordinates (in this case, Percent of Homework Completed) to divide the data set into three similarly sized groups.

Lower Group		Middle Group		Greatest Group	
Percent of Homework Completed	Unit Test Grade	Percent of Homework Completed	Unit Test Grade	Percent of Homework Completed	Unit Test Grade

8. For each group:
- Find the median of the x -values.
 - Find the median of the y -values.
 - Use these median values to write one ordered pair from each group.

L :

M :

G :

9. Add points L , M , and G to the scatter plot you constructed in Item 1.
10. Write the equation of the line containing points L and G .

My Notes

MATH TIP

When a set of numbers is listed in order, the median is the middle term. If the set contains an even number of terms, the median is the average of the two middle terms.

My Notes

11. Write the equation of the line that goes through point M and is parallel to the line that contains points L and G .

12. Compute the y -intercept of the median-median line by averaging the three y -intercept values for the lines that are associated with points L , M , and G . This averaging gives equal weight to each of the three ordered pairs of L , M , and G in the y -intercept computation. (*Note:* The y -intercept value associated with L and the y -intercept value associated with G will be the same.)

13. Write the equation of the median-median line for Ms. Windle's class data and draw this line on your scatter plot in Item 1.

14. **Reason abstractly.** Based on visual inspection of your scatter plot, does the median-median line appear to yield a reasonable, well-fitting model? Justify your response.

Check Your Understanding

15. Compare your median-median line equation with a classmate's equation. Did each of you obtain the same equation? If not, determine where you differed in your process. Do you both agree that the model is reasonable or unreasonable?

Lesson 34-1

Finding the Median-Median Line

ACTIVITY 34

continued

LESSON 34-1 PRACTICE

Model with mathematics. For each of the following data sets in Items 16–17:

- Construct a scatter plot of the data.
 - Find the equation of the median-median line.
 - Draw the median-median line on the scatter plot.
- 16.** Eighteen middle school students were asked how many hours they spent playing video games in a typical week. Then each student was asked to play a new video game that none of the other students had played before. For these 18 students, the table below gives the number of hours spent playing video games per week and the score received on the first attempt at the new game.

x = Time Spent Playing	y = Score on New Game
0	80
0	84
2	85
5	99
8	115
9	120
9	118
10	114
11	125
12	130
13	139
14	150
15	168
16	171
16	174
16	174
17	184
20	196

My Notes

My Notes

- 17.** Fifteen high school boys were measured for height (in inches) and arm span (in inches).

$x = \text{Height}$	$y = \text{Arm Span}$
66	65
67	66
68	67
68	68
68	68
69	69
69	69
69	70
69	70
71	70
71	70
71	70
72	70
73	71
75	71

- 18.** From the scatter plot you constructed in Item 16, describe the association between hours spent playing video games per week and score in terms of direction, form, and strength.
- 19.** From the scatter plot you constructed in Item 16, would a linear model be appropriate to describe the relationship between hours spent playing video games per week and score? Explain your reasoning.
- 20.** Draw a scatter plot with 20 points where a linear model would not be a good way to describe the relationship.

Using the Median-Median Line

continued

- Find the median-median line for bivariate numerical data.
- Use the median-median line to make predictions.

Use the equation for the median-median line that you wrote in Item 13 in Lesson 34-1 to answer each of the following:

- 1. Make sense of problems.** Explain the meaning of the slope of the median-median line in the context of Ms. Windle's algebra class.
- 2.** Explain the meaning of the y -intercept of the median-median line in the context of Ms. Windle's algebra class. Does this value seem reasonable in the context of these data?
- 3.** What is the risk in assuming that the pattern of the median-median line you've developed will be appropriate for homework completion percentages less than 25% (the lowest value in your data set)?
- 4.** Use the median-median line to predict the unit test score for a student who completes 72% of the homework.

My Notes

My Notes

Check Your Understanding

Write your answers on notebook paper or grid paper. Show your work.

A local car dealership recently received its shipment of a highly anticipated new model of car. Many customers needed to borrow some money in order to purchase one of these new cars. The data set below shows the amount borrowed (in dollars) by 11 customers to purchase their new car as well as the number of months that each customer had been saving up for purchasing his or her new car.

Number of Months Spent Saving for the Purchase	Amount Borrowed (in dollars)
0	21,500
1	18,000
1	17,380
2	17,500
4	14,600
5	13,958
8	12,100
10	11,000
11	9,870
16	7,300
18	5,120

- Make a scatter plot of amount borrowed to purchase the car (y -axis) in dollars versus the number of months the customer had been saving up for the car (x -axis). Does a linear model seem appropriate? Justify your response.
- Write the equation of the median-median line that estimates the amount borrowed (y) based on the number of months spent saving up for the car (x) for these 11 customers.
- Use this equation to predict the amount borrowed by a customer who had been saving up for 7 months.
- Reason abstractly.** What graphical and numerical results can be used to determine if using a linear model is appropriate? Explain.

Lesson 34-2

Using the Median-Median Line

ACTIVITY 34

continued

LESSON 34-2 PRACTICE

Here are the data on hours of TV per week and test scores from Activity 32.

x = Hours of TV per Week	y = Test Score
11	100
11	90
12	95
13	85
15	85
15	90
17	85
19	85
20	85
21	88
28	75
32	66

9. Construct a scatter plot for this data set.
10. Write the equation of the median-median line.
11. Graph the median-median line on your scatter plot.
12. Using the median-median line, what test score would you predict for someone who watches 25 hours of TV per week?
13. Using the median-median line, what test score would you predict for someone who watches 10 hours of TV per week?
14. **Construct viable arguments.** Explain why it is not a good idea to use the median-median line to predict the test score for someone who watches TV 0 hours per week or someone who watches TV 60 hours per week.

My Notes

ACTIVITY 34 PRACTICE

Use the data in the table below to complete Items 1–4.

x	4	2	1	8	3	5	9	13
y	6	14	11	5	8	2	4	3

- Construct a scatter plot for the data set.
- Write an equation for the median-median line.
- Graph the median-median line on your scatter plot.
- Do you think the median-median line is a good fit for the data? Justify your response.

The table below contains data on height (in inches) and distance jumped (in inches) for 20 high school girls. Use this table to complete Items 5–9.

x = Height	y = Distance Jumped
62	62
64	63
64	63
64	64
65	66
65	65
65	64
66	67
66	65
66	63
66	64
66	68
66	65
66	66
67	66
67	65
67	68
68	68
69	69
69	70

- Construct a scatter plot for this data set.
- Write an equation for the median-median line.
- Graph the median-median line on your scatter plot.
- Using the median-median line, how far would you predict a girl who is 66 inches tall would be able to jump?
- Do you think the median-median line is a good fit for the data? Justify your response.

The following table describes the federal minimum hourly wage since 1955. Use this table to complete Items 10–14.

# of Year	0	6	10	12	15	22	31	45	56
Hourly Wage	0.75	1.15	1.25	1.40	1.60	2.30	3.35	5.15	7.25

- Construct a scatter plot for this data set.
- Write an equation for the median-median line.
- Graph the median-median line on your scatter plot.
- Using the median-median line, predict what the federal minimum hourly wage will be in the year 2020.
- Do you think the median-median line is a good fit for the data? Justify your response.

MATHEMATICAL PRACTICES
Attend to Precision

- Chuck has made a mistake in finding the equation for a median-median line. He knows he has the correct coordinates for the points L , M , and G . The slope of his line is correct but the y -intercept is not. Which calculations should Chuck go back and check to find where he made his mistake?

Two-Way Tables and Association

ACTIVITY 35

Student Opinions

Lesson 35-1 Two-Way Tables

Learning Targets:

- Analyze two-way tables and find relative frequencies.
- Construct segmented bar graphs to display association.

SUGGESTED LEARNING STRATEGIES: Activate Prior Knowledge, Look for a Pattern, Think-Pair-Share

Prior to an upcoming school election, Greg is curious about students' opinions on some school issues. He is hoping that by carrying out a survey of the students, he will learn about student opinions before he runs for office.

Greg is particularly interested in two questions—one regarding recent changes in the food served in the cafeteria, the other regarding a recent Spirit Day held at the school. Since Greg suspects that students from different grades may feel differently about these issues, he develops a survey with the following questions:

Student Survey	(circle one answer for each)		
Do you support the new policy of healthier choices being served in the school cafeteria?	Support	Oppose	Not Sure
Do you feel the recent Spirit Day was a success in terms of raising school spirit?	Success	Not a Success	Not Sure
What grade are you in?	Seventh	Eighth	

Every student in the school completes the survey.

My Notes

My Notes

MATH TERMS

A **two-way table** is used for identifying each observation, response, etc., in a data set according to two categorical variables.

Categorical variables are used to represent categorical (as opposed to numerical) data.

Greg summarizes the data from his first question using a **two-way table** as shown below. It is called a two-way table because each student's response is identified according to two **categorical variables**.

Do you support the new policy of healthier choices being served in the school cafeteria?

	Support	Oppose	Not Sure	Total
Seventh	111	30	9	150
Eighth	180	55	15	250
Total	291	85	24	400

The table shows that there are 250 eighth graders in the school, and that 291 seventh- and eighth-grade students in the entire student body support the new policy of healthier choices being served in the school cafeteria.

1. What percentage of the seventh graders support the new policy of healthier choices being served in the school cafeteria?
2. What percentage of the eighth graders support the new policy of healthier choices being served in the school cafeteria?
3. **Reason quantitatively.** Are the percentages you computed in Items 1 and 2 similar or very different? From a percentage standpoint, does one grade seem to be much more in favor of the healthier choices being served in the school cafeteria, or are the two groups' percentages about the same in terms of supporting the healthier choices?

Lesson 35-1

Two-Way Tables

ACTIVITY 35

continued

A **segmented bar graph** is helpful for visualizing the similarities and differences between groups in a case like this. The first step to making a segmented bar graph involves computing the percentages of each response for each group of interest.

- In the table below, write the percentages you computed in Items 1 and 2. Also, compute and record the percentages of Oppose and Not Sure for each grade.

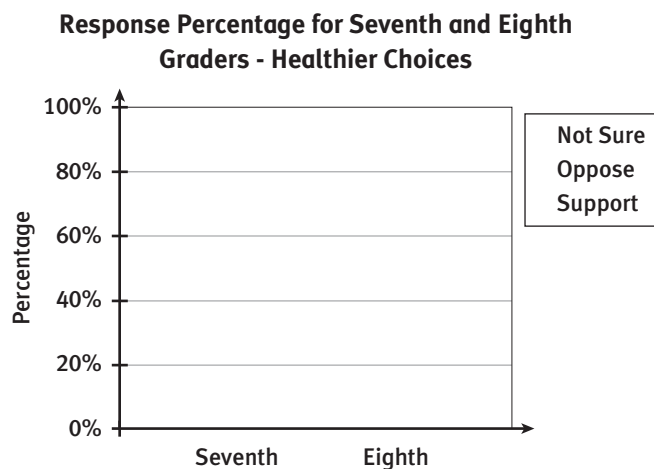
Do you support the new policy of healthier choices being served in the school cafeteria?

	Support	Oppose	Not Sure	Total
Seventh				
Eighth				

- Find the sum of the percentages in each row. These percentages are called **row percentages** (or relative frequencies) because the percentages are relative to each row category.

Based on the row percentages in the table, construct a segmented bar graph.

- Draw one bar for seventh grade and one bar for eighth grade. What is the total height of each bar?



- Model with mathematics.** Use the row percentages for each response to divide each bar into segments.
- Include a key with your graph to distinguish the segments.

Note that the order of stacking is the same for both the seventh-grade and eighth-grade bars.

My Notes

MATH TERMS

A **segmented bar graph** compares how two or more groups are divided into categories.

MATH TERMS

In a two-way table, **row percentages** communicate the relative frequencies of column categories within a given row category.

My Notes

CONNECT TO AP

In AP Statistics, you will learn formal methods for testing whether or not the two variables in a two-way table are associated.

Check Your Understanding

9. Does grade level seem to matter in terms of the response percentages?
10. In the analysis of the “healthier options” survey question above, the percentages for each response were fairly similar for each grade. If many seventh graders switched from Support to Oppose, how might the row percentages change, and how might the segmented bar graph above look different? How would the graph remain the same?

LESSON 35-1 PRACTICE

Students at a large middle school were surveyed, and each student responded to a question about after-school activities. The data are given in the two-way table below.

	Participates in After-School Activities	Does Not Participate in After-School Activities	Total
Sixth Grade	160	90	250
Seventh Grade	150	150	300
Eighth Grade	50	150	200
Total	360	390	750

11. Complete the following table by entering the missing percentages for those who do and who do not participate in after-school activities for each grade.

	Participates in After-School Activities	Does Not Participate in After-School Activities	Total
Sixth Grade			100%
Seventh Grade			100%
Eighth Grade			100%

12. How many segments will each bar have?
13. Construct a segmented bar graph for each grade that shows the percentages who do and who do not participate in after-school activities.
14. **Make use of structure.** Do the three grades appear to be different with respect to the percentages that participate in after-school activities?

Learning Targets:

- Understand association between two categorical variables.
- Describe association between two categorical variables.

SUGGESTED LEARNING STRATEGIES: Look for a Pattern, Think-Pair-Share

Like numerical variables, two categorical variables can also exhibit association. For Greg's first survey question regarding the healthier cafeteria menu, there does not appear to be an association between the response and the student's grade level.

For the second survey question regarding the Spirit Day activities, Greg makes another two-way table to summarize the results.

Do you feel the recent Spirit Day was a success in terms of raising school spirit?

	Success	Not a Success	Not Sure	Total
Seventh	96	45	9	150
Eighth	130	109	11	250
Total	226	154	20	400

1. Compute the row percentages for each grade and complete the table below.

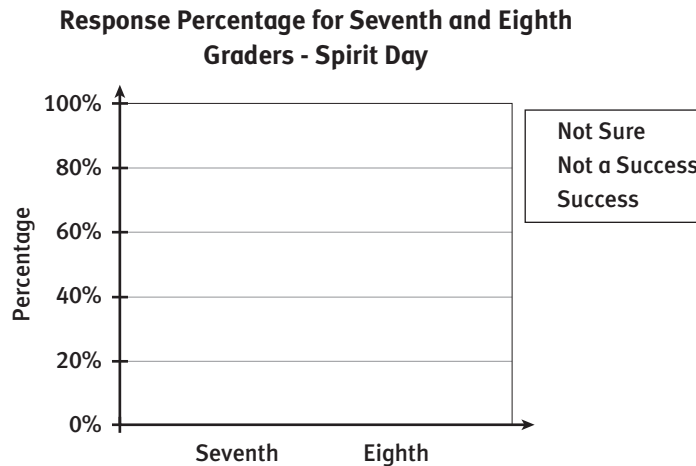
Do you feel the recent Spirit Day was a success in terms of raising school spirit?

	Success	Not a Success	Not Sure	Total
Seventh				100%
Eighth				100%

2. Based on your work in Item 1, do you think that there is an association between a student's response and his or her grade level? For example, do you think that a seventh grader is more likely (or less likely) to claim that Spirit Day was a success compared to an eighth grader? Explain your reasoning.

My Notes

3. Based on the table above, construct a segmented bar graph.



4. **Construct viable arguments.** Does your graph in Item 3 support your response to Item 2? Explain.

The following information presents the results of a survey where men and women were asked a question to which the only possible answers were Support or Oppose.

In the survey, 44% of men stated they support the new law, while only 28% of women said they support the new law.

5. **Make sense of problems.** Can you compute the overall percentage of people in the survey who answered Support from this information alone? Explain why or why not.

continued

- What are two categorical and two numerical variables that describe students at your school?
- What characteristics of a two-way table and a segmented bar graph can be used to assess if an association exists between two categorical variables? Explain.

Susan wondered if soccer players have more injuries than students who play football or basketball. She went to the local high school and asked 40 students who play soccer if they had been injured in the last year. She also asked 50 students who play football and 30 students who play basketball. Susan's data are summarized in the table below.

	Injured	Not Injured	Can't Remember	Total
Soccer	8	30		40
Football	12		3	50
Basketball	6		1	30
Total	26	88	6	120

8. Fill in the remaining cells in the table.
9. What percentage of soccer players were injured in the last year? What percentage of football players were injured?
10. Complete the table below by entering the row percentages.

	Injured	Not Injured	Can't Remember	Total
Soccer				100%
Football				100%
Basketball				100%

11. Use the percentages from Item 10 to create a segmented bar graph that compares injuries for each sport.
12. **Make use of structure.** Is there evidence of an association between whether a student is injured and the sport played? Justify your response.

ACTIVITY 35 PRACTICE

1. A football team manager took a team survey regarding where to have the team dinner after the final game of the season. The players play on either offense or defense only, and all players on the team were assigned to an offensive or defensive position.

Based on the table below, what percentage of defensive players preferred Burger Bungalow?

	Pizza Palace	Burger Bungalow
Offense	23	21
Defense	35	9

- a. 52.3%
- b. 47.7%
- c. 79.5%
- d. 20.5%

One thousand adults were asked about seat belt usage. Gender was also recorded for each person. The data are summarized in the table below.

	Never Use Seat Belt	Sometimes Use Seat Belt	Always Use Seat Belt	Total
Males	50	120	330	500
Females	75	100	325	500
Total	125	220	655	1000

2. What percentage of males always wore seat belts?
3. Complete the following table by calculating row percentages.

	Never Use Seat Belt	Sometimes Use Seat Belt	Always Use Seat Belt	Total
Males				100%
Females				100%

4. Construct a segmented bar graph using the row percentages you calculated in Item 3.
5. Does seat belt use seem to differ for males and females? If so, how does it differ?
6. Is there evidence of an association between gender and seat belt usage? Justify your response.

A class survey at a local college asked students to provide their class level and political party affiliation. Some of the results of this survey are shown in the table below.

	Political Party			
	Democrat	Republican	Other	Total
Freshman	1		1	6
Sophomore	4	8		15
Junior		4	3	12
Senior	3		2	7
Total	13	18	9	40

7. Complete the two-way table.
8. What percentage of sophomores in the class affiliate themselves as “Other”?
9. What percentage of juniors in the class affiliate themselves as “Democrat”?
10. Calculate row percentages to complete the table below.

	Political Party			
	Democrat	Republican	Other	Total
Freshman				100%
Sophomore				100%
Junior				100%
Senior				100%

11. Construct a segmented bar graph using the row percentages you calculated in Item 10.
12. Is there evidence of an association between class level and political party affiliation? Justify your response.

MATHEMATICAL PRACTICES

Create Viable Arguments and Critique the Reasoning of Others

13. Chuck’s group is creating a segmented bar graph involving sixth, seventh, and eighth graders. Since there are more seventh graders, Chuck believes the group should begin with the bar representing the seventh grade because that bar will be the tallest. Do you agree or disagree with Chuck? Explain why.

Median-Median Line and Two-Way Tables

MOKHER'S MEASUREMENTS

Embedded Assessment 2

Use after Activity 35

1. After taking a random sample of 12 children from Grade 0 (Kindergarten) through Grade 6, Mr. Mokher and his students collected data on each child's grade level and weight in pounds.
 - a. Construct a scatter plot of weight (y -axis) versus grade (x -axis).
 - b. Describe the overall association between grade level and weight in pounds in terms of direction, form, and strength. If necessary, identify any observations that substantially deviate from the overall pattern.
 - c. Write the equation of a median-median line for this relationship that could be used to predict weight based on grade level. Graph the line on your scatter plot and comment on how well the model seems to fit the data.
 - d. Interpret the slope of the median-median line in the context of this problem.
 - e. Interpret the y -intercept of the median-median line in the context of this problem.
 - f. Using the median-median line, what is the predicted weight of a third grader?
2. A survey conducted by the Pew Research Center in April–May 2010 as part of their “Internet & American Life Project” obtained the following results based on a sample of individuals:

Student	Grade	Weight (lbs.)
1	0	39
2	0	40
3	0	42
4	1	45
5	2	47
6	2	52
7	3	57
8	4	62
9	5	68
10	5	72
11	5	63
12	6	75

Survey Question: Do you use the Internet, at least occasionally?

Age	Yes	No	Total
18–29	1,845	143	1,988
30–49	2,459	1,657	4,116
50–64	937	508	1,445
65 and older	559	810	1,369
Total	5800	3,118	8,918

- a. Calculate the row percentages for each row category above. Then create a table showing the row percentages. Round your percentages to the nearest tenth of a percent.

- b. Create a segmented bar graph that compares the Yes and No response percentages for each age group based on the row percentages above. Be sure to include a key.
- c. Is there evidence of an association between age and response to the question? Justify your response, referencing your work in parts a and b in your explanation.

Scoring Guide	Exemplary	Proficient	Emerging	Incomplete
	The solution demonstrates these characteristics:			
Mathematics Knowledge and Thinking (Items 1a-f, 2a-c)	Clear and accurate understanding of scatter plots, median-median lines, and other data displays.	A functional understanding of scatter plots, median-median lines, and other data displays.	Partial understanding of scatter plots, median-median lines, and other data displays.	Little or no understanding of scatter plots, median-median lines, and other data displays.
Problem Solving (Items 1f, 2c)	Clear and accurate interpretation of data displays to make a prediction and to characterize an association.	Interpreting data displays to make a prediction and to characterize an association.	Difficulty making an accurate prediction or characterizing an association using data displays.	Inaccurate interpretation of data displays.
Mathematical Modeling / Representations (Items 1a, c, 2a-b)	Clear and accurate understanding of creating tables and graphs from data.	Creating tables and graphs from data that are mostly correct.	Errors in creating tables and graphs from data.	Inaccurate or incomplete tables and graphs
Reasoning and Communication (Items 1b-f, 2c)	Precise use of appropriate math terms and language to characterize associations in bivariate data.	Correct characterization of associations in bivariate data.	Misleading or confusing characterization of associations in bivariate data.	Incomplete or inaccurate characterization of associations in bivariate data.